

What is Phonetics?

- **Definition:**

- The study of speech sounds, their production, transmission, and perception.

- **Three Main Branches:**

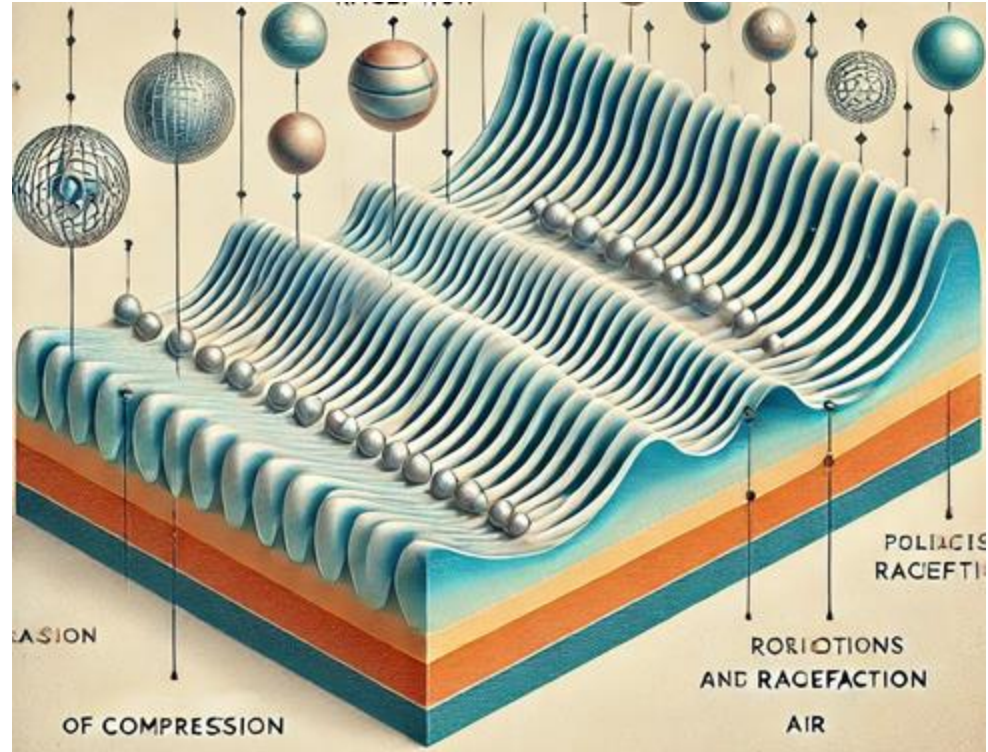
- **Articulatory Phonetics:** How speech sounds are produced
- **Acoustic Phonetics:** Physical properties of speech sounds
- **Auditory Phonetics:** How speech sounds are perceived by the listener

Example: Different between /b/ and /p/ sounds



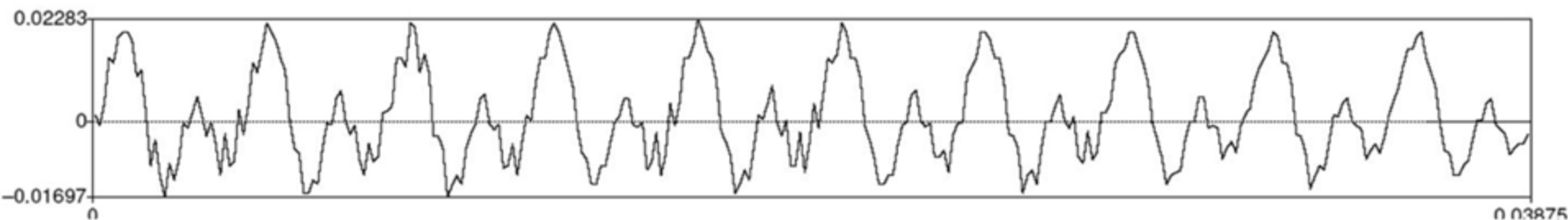
Sound Waves are Longitudinal Waves

- Particles of the medium vibrate parallel to the direction of the wave's propagation
 - e.g. Sound waves, P-waves, pressure waves....
- Repeating patterns described by **wavelength (bước sóng)**, **frequency**, **amplitude**, and **speed**



Back to Waves: Fundamental Frequency

- Waveform of the vowel /iy/



- Frequency: $10 \text{ repetitions} / 0.03875 \text{ seconds} = 258 \text{ Hz}$
- This is speed that vocal folds move, hence voicing
- Each peak corresponds to an opening of the vocal folds
- The low frequency of the complex wave is called the fundamental frequency of the wave or F0

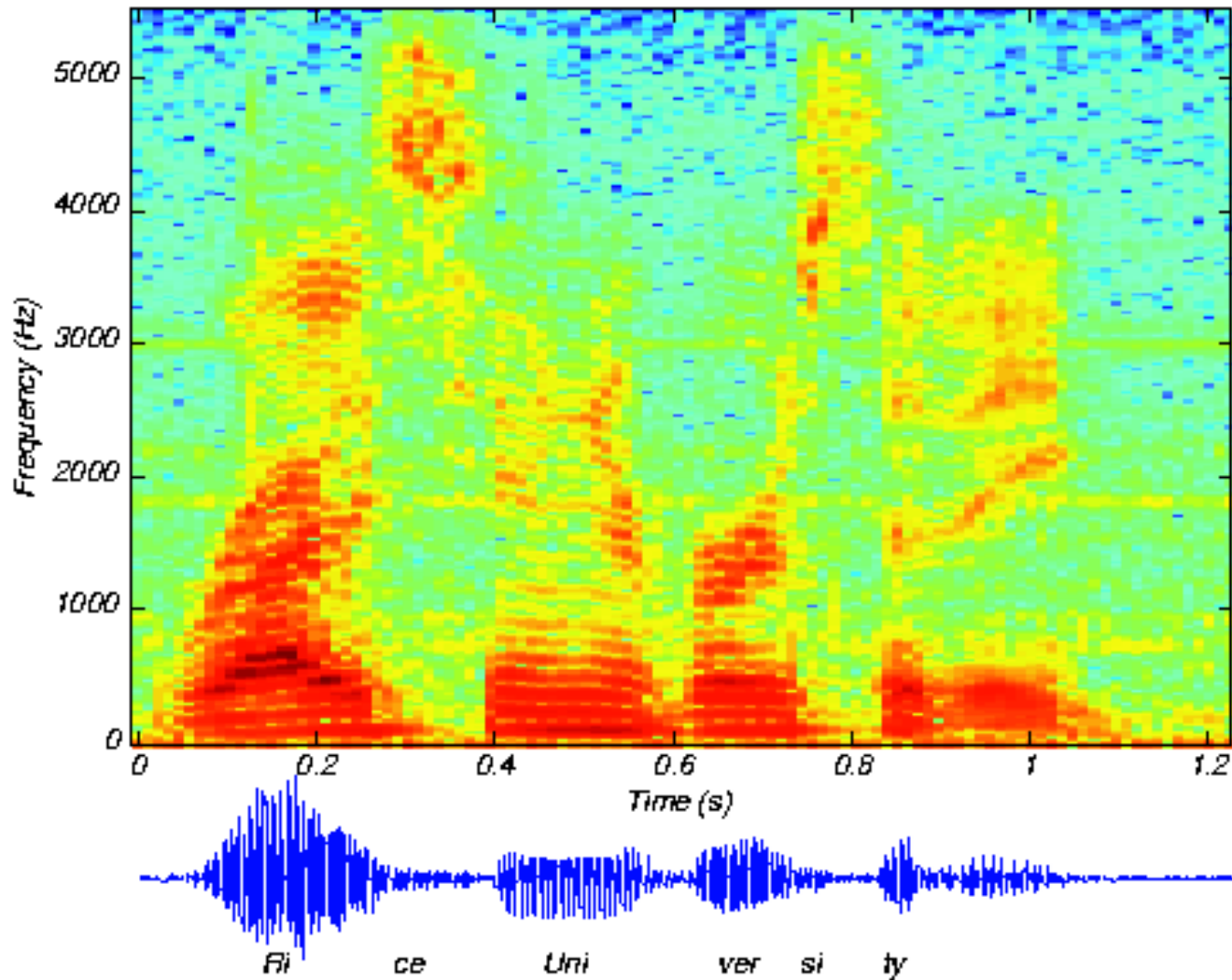
Fundamental Frequency

Fundamental Frequency



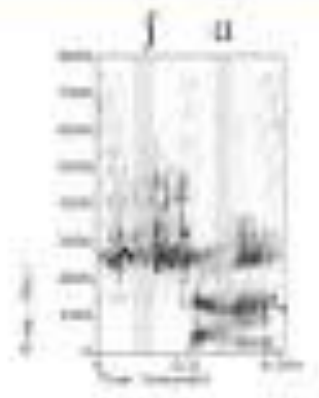
Spectrogram (Ảnh phổ)

A graph that displays the **strength** of a **signal** over **time** for a given **frequency** range



Waveforms and Spectrograms

Waveforms and Spectrograms



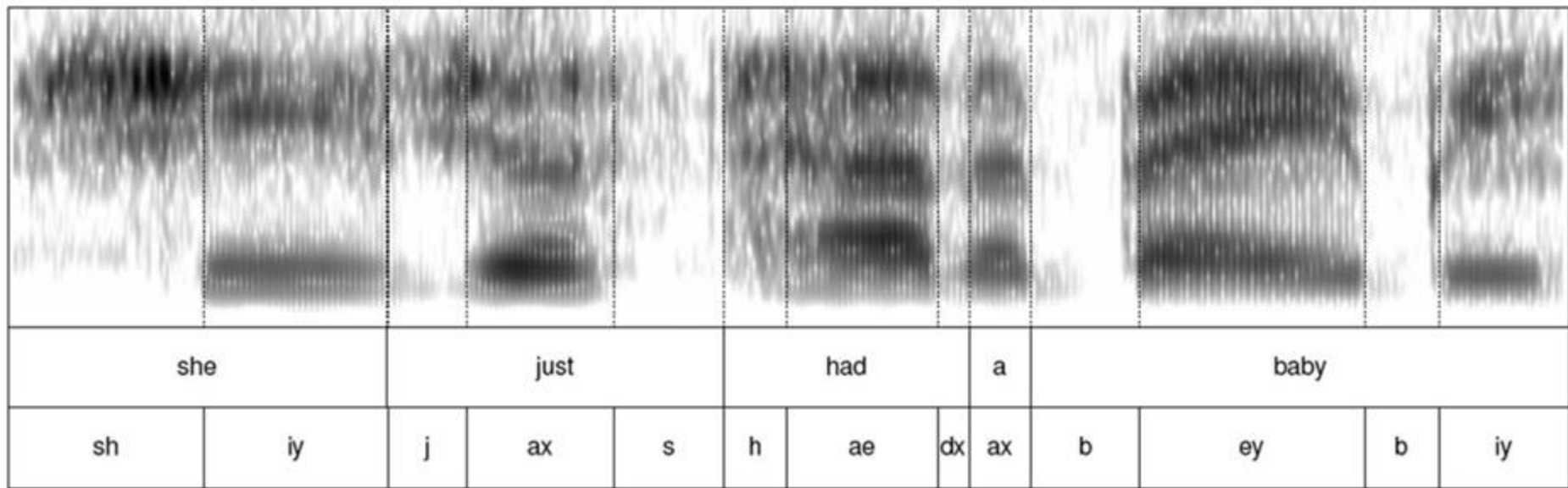
Spectrogram: Spectrum + Time Dimension

SHE

JUST

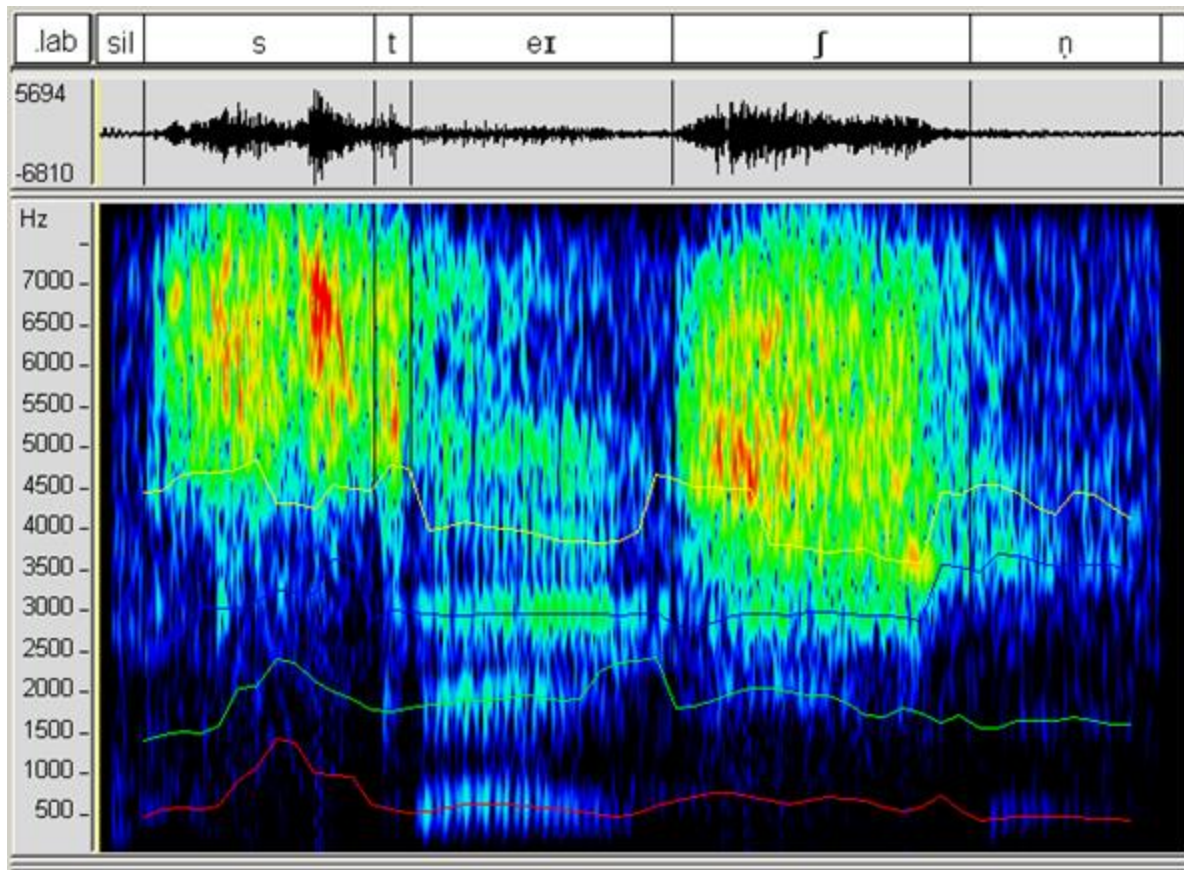
HAD

A



Consonant and Vowel Spectrograms

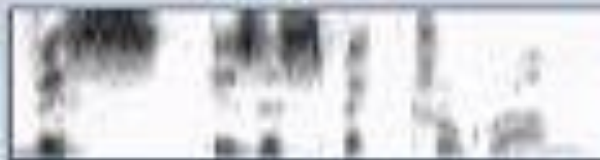
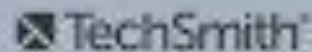
- **Vowels** have **full formant patterns**, based mostly on F1 and F2 and F3
- **Consonants**: No **clear formant patterns** themselves, but rather affect the patterns of the vowels surrounding them



How to read vowel spectrograms?



Reading Spectrograms a VLC Series by Prof. Dr. Jürgen Handke



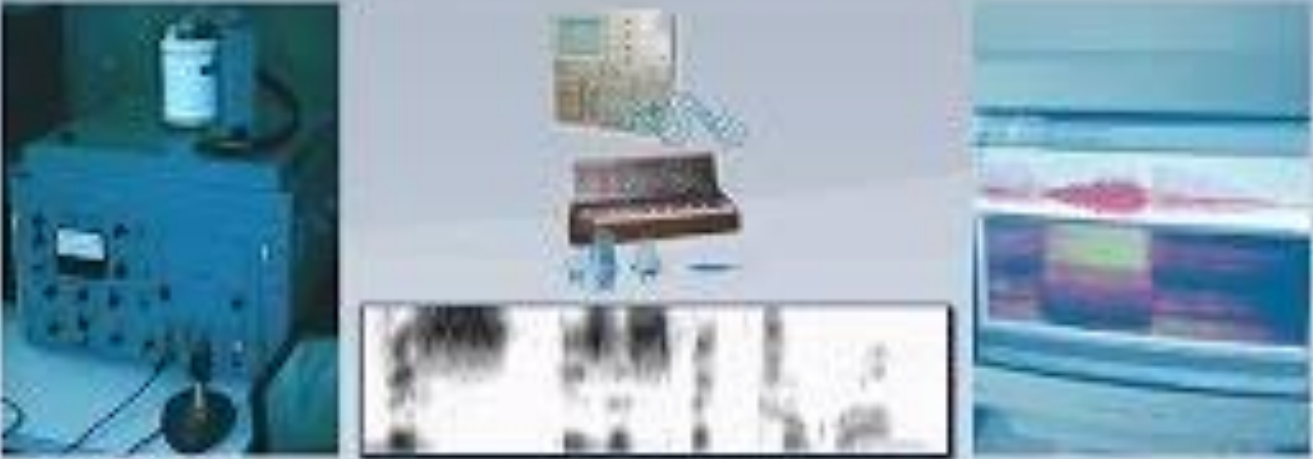
Vowels

How to read consonant spectrograms?

The Virtual Linguistics Campus

Reading Spectrograms
a VLC Series
by Prof. Dr. Jürgen Handke

TechSmith®



Consonants

Reading Spectrograms: Consonants © The Virtual Linguistics Campus, Marburg University, Germany

Review: Grapheme vs. Phoneme

- **Grapheme**

The smallest written unit representing a sound in speech.

Examples: "d", "th", "ngh", "a", "au".

- **Phoneme**

The smallest unit of sound in speech.

Examples: /d/, /th/, /ŋ/, /a/, /ăw/.

Review: IPA (International Phonetic Alphabet)

BẢNG PHIÊN ÂM TIẾNG ANH IPA

VOWELS	monophthongs				diphthongs		Phonemic Chart	
	i:	ɪ	ʊ	u:	ɪə	eɪ		
	leave	live	put	pool	year	paper		
	e	ə	ɜ:	ɔ:	ʊə	ɔɪ		
pen	a	girl	door	tourist	boy	phone	voiced unvoiced	
æ	ʌ	a:	ɒ	eə	aɪ	aʊ		
man	cup	car	hot	chair	fine	house	Designed by HapTOEC	
CONSONANTS	p	b	t	d	tʃ	dʒ	k	g
	pen	baby	table	door	chip	January	key	girl
	f	v	θ	ð	s	z	ʃ	ʒ
	fan	van	think	this	sun	zoo	shoes	vision
m	n	ŋ	h	l	r	w	j	
mouth	nose	ring	hear	letter	rain	window	yellow	

leave /li:v/
manly /'mænli/

Exercise

- Download Praat: www.praat.org
- Record the sentence: "She just had a baby"
- Create 2 tiers with TextGrid for annotating for the word (grapheme) and phoneme (using IPA) levels

Popular packages for signal analysis

- Praat (www.praat.org)
 - by Paul Boersma & David Weenink, Phonetics Amsterdam
- Wavesurfer (www.speech.kth.se/wavesurfer/)
 - by scientists at Stockholm Technical University - KTH
- Goldwave (www.goldwave.de)
 - commercial program, but available free for trial use

2.2 Vietnamese Phonology and Phonetics



Nguyen Thi Thu Trang

A traditional Vietnamese painting depicting a serene landscape. In the center, a multi-tiered pagoda with a brown roof stands on a small, grassy island in a calm body of water. The pagoda is surrounded by lush green trees and foliage. In the foreground, a dense field of lotus flowers with large green leaves and pink blossoms is visible. The scene is framed by dark, gnarled tree branches with small green leaves at the top and sides. The overall style is characteristic of traditional Vietnamese art, with a focus on nature and architecture.

**Why is Vietnamese
interesting?**

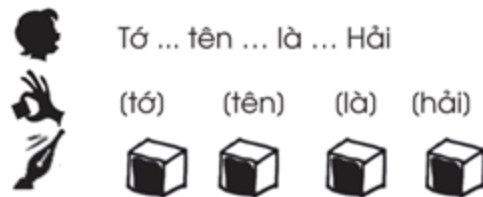
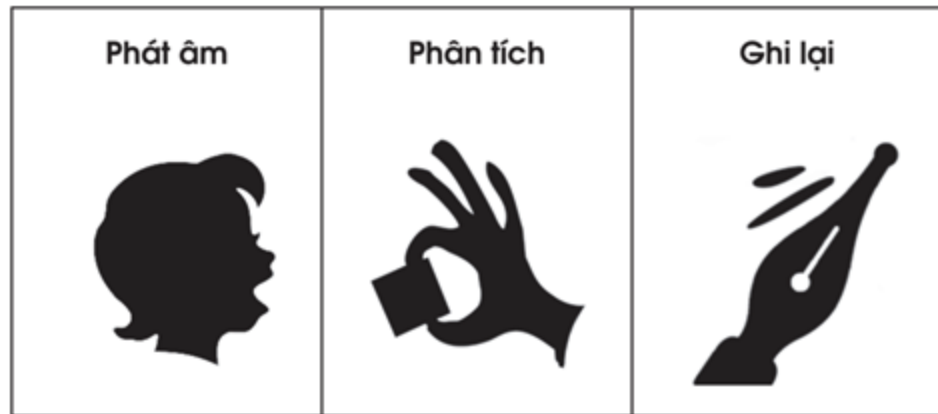
What is Vietnamese?

- Mon-Khmer sub-branch of the Austroasiatic (Nam Á) language family with mixture of Tai-Kadai (Tày-Thái)
- Official language of Vietnam
- ~100 millions speakers in the world



Tiếng Việt là ngôn ngữ đơn lập (monosyllabic)

- Tiếng/âm tiết (syllable) được phát âm và viết tách rời
 - Công ty vs. Company
- Từ (word) của tiếng Việt không biến đổi hình thái
 - Tôi **đã** xem phim này rồi.
- Trật tự từ quan trọng
 - Con gà vs. Gà con.



Tiếng Việt là ngôn ngữ có thanh điệu (tonal)

1 - Ngang (A1)

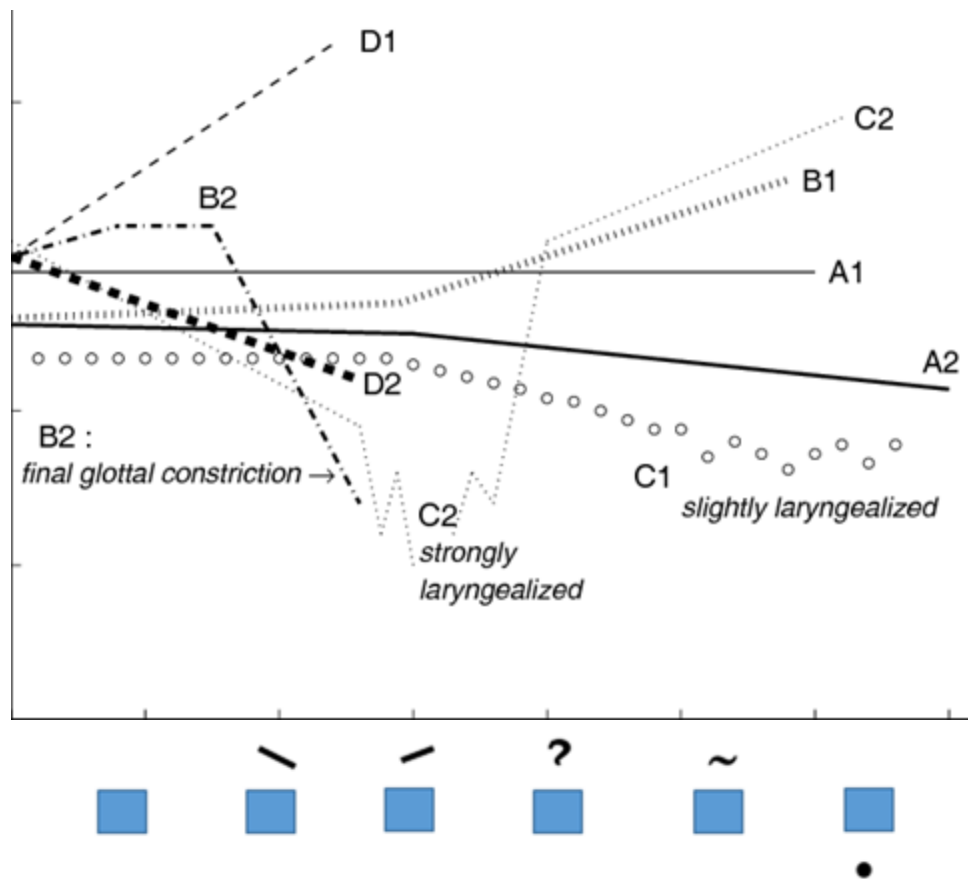
2 - Huyền (A2)

3 - Hỏi (C1)

4 - Ngã (C2)

5a, 5b - Sắc (B1, D1)

6a, 6b - Nặng (B2, D2)



Hanoi Vietnamese¹ tone system *(Michaud 2004)*

- 6 tones in sonorant-final syllables
- 2 tones in obstruent-final syllables (ending with /p t k/)

#	Name	Duration	Example
1	Level	Long	ba [ʃa-1] (three)
2	Falling	Long	bà [ʃa-2] (grand-mother)
3	Curve	Long	bả [ʃa-3] (poison)
4	Broken	Long	bã [ʃa-4] (residue)
5a	Rising	Long	bá [ʃa-5a] (to embrace)
5b	Rising*	Short	bác [ʃa-5b] (uncle)
6a	Drop	Short	bạ [ʃa-6a] (to strengthen)
6b	Drop*	Short	bạc [ʃa-6b] (silver)

¹*Standard Vietnamese (Thompson 1987)*

Michaud (2004). Michaud Alexis. *Final consonants and glottalization: new perspectives from Hanoi Vietnamese*. *Phonetica*.

Thompson (1987). Thompson Laurence C. *A Vietnamese Reference Grammar*. University of Hawaii Press.

Hanoi Vietnamese - G2P Rules for consonants

No.	Graph-eme	Position	Pho
			IPA
1	b	initial only	ɓ
2	đ	initial only	ɗ
3	x, s	initial only	s
4	g, gh	initial only	ɣ
5	kh	initial only	x
6	l	initial only	l
7	v	initial only	v
8	th	initial only	t ^h
9	d, gi, r	initial only	z
10	ph	initial only	f
11	tr, ch	initial only	tʃ
12	h	initial only	h
13	c, k, q	initial only	k

No.	Graph-eme	Position	Pho
			IPA
14	t	initial, final	t
15	p	initial, final	p
16	n	initial, final	n
17	ch	final after i, ê, a	k
18	c	final after u, o, ô	k ⁺ p
19	ch, c	final except 17, 18	k
20	m	initial, final	m
21	nh	final after i, ê, a	ŋ
22	nh	initial only	ɲ
23	ng, ngh	initial	ŋ
24	ng	final after u, o, ô	ŋ [̃] m
25	ng	final except 24	ŋ

Hanoi Vietnamese - G2P Rules for vowels

Vowel type	Grapheme (Orthography)	Phoneme
		IPA
Long vowel	a	a
	e	ɛ
	ê	e
	i, y	i
	o, oo	ɔ
	ô, ôô	o
	ơ	ɤ
	u	u
	ư	ʊ

Vowel type	Grapheme (Orthography)	Phoneme
		IPA
Short vowel	ă, a (au, ay)	ă
	â	ɤ
	a (anh, ach)	ɛ
	o (ong, oc)	ɔ
Diphthong	ia, iê, yê, ya	iə
	ua, uô	uə
	ưa, ươ	ʊə

Vietnamese syllable structure

- Hierarchical structure

(Doan 1977) (Vogel 2004)

- **Segmental** elements

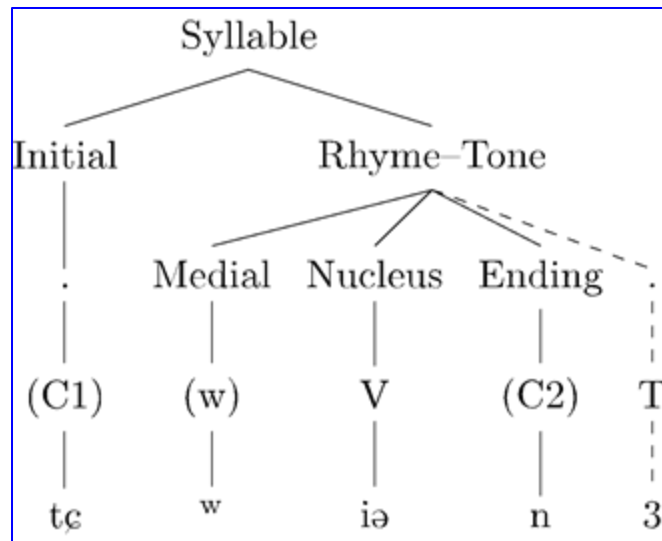
- Initial consonant
- Rhyme (medial, nucleus, ending)

- **Supra-segmental** element:

Lexical Tone (T)

- mostly covers the rhyme of syllables (Tran 2005)

- Syllabic structure: (C1)(w)V(C2)-T



E.g. chuyên [tɕ²wiən-3] (to move)

Doan (1977). Doan Thien Thuat. *Vietnamese phonetics (in Vietnamese)*. Vacation School and University Publisher.

Doan (1999). Doan Xuan Kien. *Re-consider a problem of Vietnamese phonetics: Syllable structure (in Vietnamese)*. Hop Luu

Tran (2005). Tran Do Dat et al. *Influence of F0 on Vietnamese Syllable Perception*. Interspeech 2005. Portugal.

Vogel (2004). Vogel Irene et al. *Syllable structure in Vietnamese*. TEAL Workshop. Taiwan.

Structure-based syllable types

INITIAL	RHYME			Examples
	Medial	Nucleus	Ending	
		v		“a” /a-1/ (<i>ah</i>), “ủ” /ɤ-3/ (<i>keep wa</i>
v		v		“lá” /la-5a/ (<i>leaf</i>), “chờ” /tɕɤ-2/ (<i>wa</i>
	v	v		“oẹ” /we-6a/ (<i>retch</i>), “uy” /wi-1/ (<i>wa</i>
		v	v	“ích” /ik-5b/ (<i>helpful</i>), “ây” /ɤ̃-5a/ (<i>wa</i>
v	v	v		“loé” /lwe-5a/ (<i>flash</i>), “quá” /kwa-6a/ (<i>wa</i>
v		v	v	“trẹo” /tɕew-6a/ (<i>sprain</i>), “nhãn” /ɲwien-6a/ (<i>wa</i>
v	v	v	v	“nhuyễn” /ɲwien-4/ (<i>fine</i>), “soát” /sɔt-6a/ (<i>wa</i>

Allophone – []

- A variant of a phoneme
- The allophones of a phoneme form a set of sounds that:
 - Do not change the meaning of a word
 - Are all very similar to one another, and
 - Occur in phonetic contexts different from one another (for example, syllable-initial as opposed to syllable-final).
- The differences among allophones can be stated in terms of phonological rules.

Luật phiên âm tiếng Việt - Âm đệm

- Duy nhất 01 âm đệm **/w/** được biết bằng **“o”** hoặc **“u”**
- **“o”** đứng trước các nguyên âm rộng: a, ă, e
 - VD: toát, hoạ hoản, lòe xoè
- **“u”** đứng trước các nguyên âm: y, ê, ơ, â
 - VD: huy, thuở, huân, Huế

KHUYÊN /xwiən-1/			
kh /x/	u /w/	yê /iə/	n /n/
	1		

Exercise

- Phonology differences of Northern, Central and Southern Vietnamese
- Reference:
https://www.researchgate.net/profile/Deborah-Hwa-Froelich/publication/278524880_Characteristics_of_Vietnamese_Phonology/links/02e7e530f680f7361f000000/Characteristics-of-Vietnamese-Phonology.pdf

VD: Luật chính tả tiếng Việt cơ bản

Không tính các từ vay mượn

- Âm đầu **g/gh, ng/ngh, c/k**: Trước âm chính:
 - **Hẹp/hơi hẹp**: **i, ê, e** → **gh, ngh, k**
 - VD: kê, kia, nghe, ghe, nghênh, ké, ki bo, nghề nghiệp
 - **Rộng/hơi rộng**: **a, ă, â, o, ô, ơ, u, ư** → **g, ng, c**
 - VD: cá, ngốc, gặt gù, ngừ, cố, gằm, ngủ, gặt, ngọt ngào
- Âm đầu **q/c/k** (đều có âm là /k/)
 - **q** luôn bao giờ cũng đi với âm đệm **u** để thành qu
 - VD: qua, quanh quẩn (*làm tròn môi âm chính*)
 - **c/k** như trên
- Âm cuối **c/ch** (đều có âm là /k/): Sau âm chính:
 - **Hẹp/rộng**: **i, ê, a** → **ch, còn lại là "c"**
 - VD: bích, ốc, ếch, khách, ích, lúc, bác, bức xúc

Exercise - Luật chính tả / phiên âm

- “gia” và “ghia”; “củ”, “cả”, “cua” và “qua”
- Quốc gia Qata, tỉnh Đắk Lắk



Discussion - Luật chính tả / phiên âm

- “gia” và “ghia”, “cua” và “qua”

gi		a	
----	--	---	--

gh		ia	
----	--	----	--

c		ua	
---	--	----	--

q	w	a	
---	---	---	--

- Quốc gia Qata, tỉnh Đắk Lắk

→ Từ vay mượn: Qatar và Dak Lak (Darlac)

Luật chính tả tiếng Việt cơ bản

Không áp dụng cho các từ vay mượn

- Âm chính: Nguyên âm y/i
 - Viết là “y”
 - Sau âm đệm “u”: huy, tuy, thúy, quy...
 - Sau nguyên âm ngắn a như: ây, ay
 - Đứng trước ê khi chữ đó không có âm đầu như: yêu, yết, yếm
 - Đứng một mình: y tế, ý nghĩ. Ngoại lệ: í ới
 - Viết “i/y” đều đúng:
 - Không có âm cuối, khuyến khích viết i: Châu Mỹ/Châu Mỹ, Địa lí/Địa lý, Bác sĩ/Bác sỹ
 - Còn lại là “i”
- Exercise
 - “bay” và “bai”, “tuy” và “tui”

Discussion - Luật chính tả / phiên âm

- “bay” và “bai”, “tuy” và “tui”

b		a	y
/b		ă	j/

t	u	y	
/t	w	i	/

b		a	i
/b		a	j/

t		u	i
/t		u	j/

Exercise

- Record in Praat for the following sentences:
“Dùng tay vuốt mái tóc bạch kim của em qua tai”
- Analyze and label at allophone level for those sentences